

New from old via logical argument

Ex:

If we know $p \rightarrow q$

and we know $p \vee q$

can we conclude anything about q

Alternatively

Using rules of inference (and logical equivalence)

Obstacle: We have a mix of disjunctions (computer think)

and implications (human thought)

Approach 1

Turn the disjunction into an implication,

1. $p \vee q$ given
2. $\therefore \neg p \rightarrow q$ (logically equivalent to 1.)
3. $p \rightarrow q$ also given
4. $(p \rightarrow q) \wedge (\neg p \rightarrow q)$ (conjunction of 2. and 3.) (conjunction rule from table of inference)
5. $(p \vee \neg p) \rightarrow q$ logically equivalent to 4.
6. $T \rightarrow q, p \vee \neg p \equiv T$
7. T absolute fact
8. q modus ponens 6. + 7.

Approach 2

Turn the implication into a disjunction

1. $p \rightarrow q$ (given fact)
2. $\neg p \vee q$ (CD equivalent to 1.)
3. $p \vee q$ (given fact)
4. $q \vee q$ (resolution rule)
5. q (equivalent to 4. via idempotency)

Fallacy

We know how to test if:

$$p_1$$
$$p_2$$
$$\dots$$
$$p_n$$

$$\therefore q$$

The above is valid,

If it isn't, then some combination of variable values demonstrates this, (rough work involves a table, the explanation is just one set of values)

Common fallacies

$$\frac{p \rightarrow q}{\therefore p}$$

False when $p = F$ and $q = T$ (Affirming the conclusion)

$$\frac{p \rightarrow q}{\therefore \neg p}$$

$$\frac{\neg p}{\therefore \neg q}$$

False when $p = F$ and $q = T$ (denying the hypothesis)

$$\frac{p \vee q}{\therefore \neg q}$$

$$\frac{p}{\therefore \neg q}$$

false when $p = T$ and $q = T$ (affirming the disjunction)

Rules involving Quantifiers

Examples:

Suppose we learn about some predicate $P(x)$ that $\forall x > 0 (P(x) \rightarrow P(x + 1))$
and also that $P(72)$ is true

$P(x)$ could be “x people is enough to form an Aussie rules football team”

We can conclude $P(74)$ is enough people to play Aussie rules football

1. $\forall x > 0, (P(x) \rightarrow P(x + 1))$ given
2. $P(72) \rightarrow P(73)$ (3. & 4. are Universal Instantiation of 1.)
3. $P(73) \rightarrow P(74)$
4. $P(72) \rightarrow P(74)$ (Hyp Syllogism of 2. and 3.)
5. $P(72)$ (given)
6. $P(74)$ (Modus Ponens on 4. and 5.)

Example

Suppose two predicates

$W(x)$ and $S(x)$ are related $\forall x (W(x) \rightarrow S(x))$

If we learn $W(x)$ is grass, then the following is valid:

1. $\forall x (W(x) \rightarrow S(x))$
2. $W(\text{grass}) \rightarrow S(\text{grass})$ (Universal Instantiation of 1.)
3. $W(\text{grass})$ (given)
4. $\therefore S(\text{grass})$ (Modus Ponens on 2. and 3.)

(1. through 4. is a common pattern, that gets its name)

1. $\forall x (W(x) \rightarrow S(x))$ (given)
2. $W(\text{grass})$ (given)
3. $\therefore S(\text{grass})$ (universal Modus Ponens on 1. and 2.)

Similarly we have Universal Modus Tollens

Sam Shortenning applies on previous argument:

1. $\forall x \geq 0 (P(x) \rightarrow P(x + 1))$
2. $P(x)$
3. $P(73)$ UMP on 1. and 2.
4. $p(74)$ UMP on 1. and 3.

1.7 & 1.8 Proofs

Terms

- Theorem $\leftarrow$ Important
- Proposition $\leftarrow$ Neutral Word
- Lemma $\leftarrow$ Stepping Stone
- Corollary $\leftarrow$ Easy consequence

$\wedge$ All things we have proved

- Conjecture $\leftarrow$ We don't know if it's true
- Many important theorems are in the form of an implication or disjunction

Ex:

If m and n are even integers, then so is $m + n$

- $\wedge$ Implied universal quantifier
 - Means "For all integers m and n , if m is even, and n is even, $m + n$ is even"

"Every integer is either even or odd,"

Definition An integer is even, iff there exists an integer k such that $n = 2k$,

For current discussion we'll write $\mathcal{E}(n)$

An integer n is odd iff $\exists k(n = 2k + 1)$